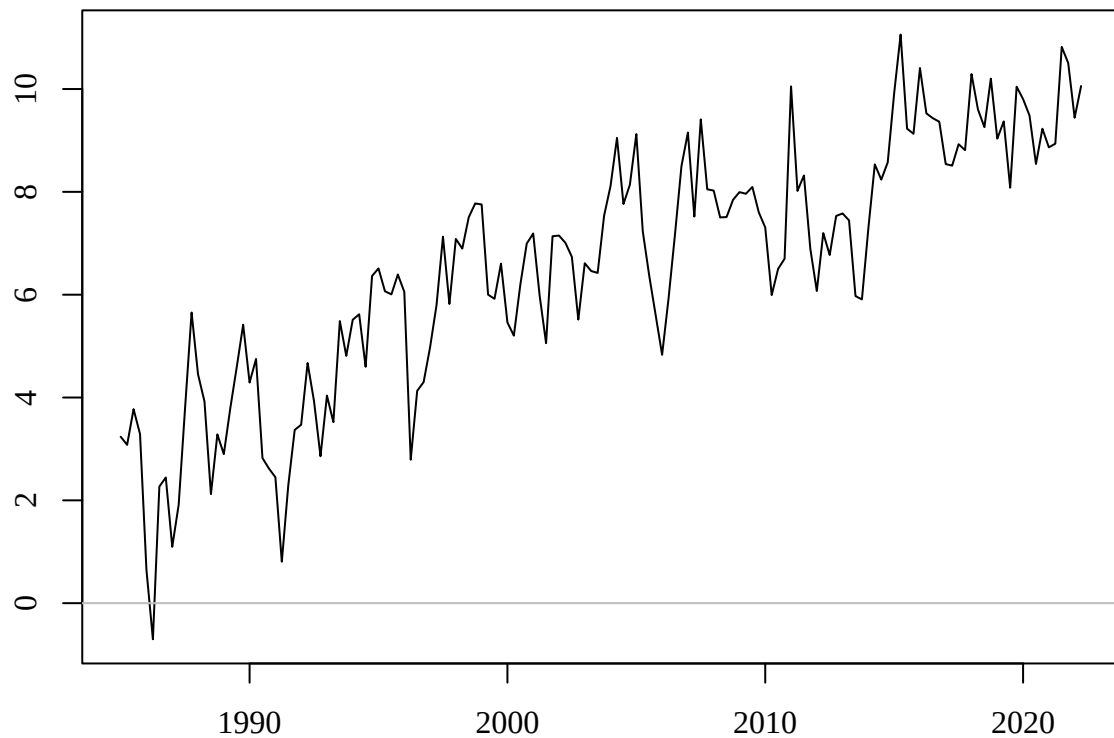


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Question 1. (8 marks)

The following is a plot of a quarterly time series from 1985q1 to 2022q2, a total of 150 observations.



- (a) (2 marks) Does this time series plot suggest a **trend** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(1 mark) Yes.

(1 mark) The plot suggests a trend because the time series shows an overall upwards movement during this time.

(b) (2 marks) Does the time series plot suggest a **seasonal pattern** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(1 mark) No.

(1 mark) The plot does not suggest any clear repeating seasonal pattern.

(c) (2 marks) Does the time series plot suggest **non-stationarity** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(1 mark) Yes.

(1 mark) The presence of the trend suggests the mean of the time series is increasing over time, which implies non-stationarity.

(d) (2 marks) Does the time series plot suggest **non-invertibility** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(1 mark) Can’t tell.

(1 mark) We cannot tell whether a moving average component is present, nor what inverse root(s) it may have.

Question 2. (11 marks)

An augmented Dickey-Fuller test regression with trend was calculated for the time series in the plot in Question 1 using the `urca` command `ur.df`:

```
ur.df(Y, type="trend", lags=1)
```

The following test regression was obtained.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.270	0.270	4.702	0.000
z.lag.1	-0.430	0.076	-5.673	0.000
tt	0.021	0.004	5.043	0.000
z.diff.lag	0.040	0.083	0.475	0.636

(a) (2 marks) Write out the estimated test regression in equation form.

$$\Delta \hat{Y}_t = 1.270 - 0.430Y_{t-1} + 0.021t + 0.040\Delta Y_{t-1}$$

(b) (1 mark) What is the value of the augmented Dickey-Fuller t test statistic?

$$t = -5.673$$

(c) (2 marks) What are the null and alternative hypotheses of the Dickey-Fuller test?

$H_0 : Y_t$ has a unit root, or $H_0 : \varphi = 0$

$H_1 : Y_t$ does not have a unit root, or $H_1 : \varphi < 0$

If the latter, define $\Delta Y_t = \beta_0 + \varphi Y_{t-1} + \beta_1 t + \psi_1 \Delta Y_{t-1} + U_t$.

The `urca` command `qunitroot(q, trend="ct")` is used to compute the critical values for the augmented Dickey-Fuller test for $q = 0.05, 0.01, 0.005, 0.001$. The output is shown here:

	5%	1%	0.5%	0.1%
ADF critical value:	-3.41	-3.958	-4.164	-4.596

- (d) **(3 marks)** Carry out the augmented Dickey-Fuller test using the statistic in part (b) using the 5% level of significance. State the test decision and conclusion.

(2 marks) $t = -5.673$ is less than the 5% critical value -3.41 , therefore reject H_0 .

(1 mark) The conclusion is that Y_t does not have a unit root.

- (e) **(3 marks)** If we were to compute the p -value for the test statistic in part (b), would this p -value be greater than or less than 0.001? Explain.

(1 mark) The p -value would be less than 0.001.

(1 mark) This is because the t statistic -5.673 is less than the 0.1% (i.e. 0.001) critical value -4.596 .

(1 mark) This in turn implies it is further into the lower tail of the distribution than the 0.001 quantile.

If you run out of space for Question 2, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ Please tick this box if you used extra pages for any part of Question 2.

Question 3. (23 marks)

For a stationary time series with $n = 200$ observations a range of $\text{ARMA}(p, q)$ were fitted to attempt to find a suitable forecasting model. The following table shows the values of the AICc and p -values of the Ljung-Box residual autocorrelation test for every combination of $p = 0, 1, 2, 3$ and $q = 0, 1, 2, 3$.

p	q	AICc	LBp
0	0	240.81	0.000
1	0	99.76	0.008
2	0	99.95	0.725
3	0	101.32	0.608
0	1	129.60	0.000
1	1	99.84	0.937
2	1	101.36	0.910
3	1	103.44	0.609
0	2	103.75	0.313
1	2	101.20	0.997
2	2	103.30	0.948
3	2	105.33	0.667
0	3	101.86	0.915
1	3	103.29	1.000
2	3	105.44	0.950
3	3	99.98	0.723

- (a) (3 marks) According to the AICc alone, what are the best three models? (List in order of first, second, third best.) Explain.

(1 mark) Smaller AICc values suggest better fitting models.

(2 marks) The AICc results suggest the best three models are, from first to third, ARMA(1,0), ARMA(1,1) and ARMA(2,0).

(b) (5 marks) Taking into account the Ljung-Box test results in addition to the AICc from part(a), which model would be selected as overall “best”? Explain how and why the Ljung-Box results are relevant to this selection.

(1 mark) The Ljung-Box p -value is less than 0.05 for the AICc-best model, ARMA(1,0), implying the null hypothesis of no residual autocorrelation is rejected for this model.

(2 marks) This model therefore not considered valid because residual autocorrelation in the residuals / prediction errors is a sign that the conditional expectation has been misspecified.

(2 marks) The second best model by the AICc has a Ljung-Box p -value greater than 0.05, so this model is valid and so the ARMA(1,1) is considered overall “best”.

The coefficient estimates of the ARMA(1,1) and ARMA(2,0) models are as follows:

	ar1	ma1	intercept
ARMA(1,1):	0.523	0.355	2.751

	ar1	ar2	intercept
ARMA(2,0):	0.874	-0.263	2.747

(c) (4 marks) Write out both models in equation form.

(2 marks) ARMA(1,1) : $Y_t = 2.751 + \hat{Z}_t$
 $\hat{Z}_t = 0.523Z_{t-1} + 0.355U_{t-1}$

(2 marks) ARMA(2,0) : $Y_t = 2.747 + \hat{Z}_t$
 $\hat{Z}_t = 0.874Z_{t-1} - 0.263Z_{t-2}$

Forecasts from the ARMA(1,1) and ARMA(2,0) models for $h = 1, 2, \dots, 8$ steps ahead are tabulated here:

h	ARMA(1,1)	ARMA(2,0)
1	3.165	3.089
2	2.967	2.851
3	2.864	2.748
4	2.810	2.720
5	2.781	2.723
6	2.767	2.733
7	2.759	2.741
8	2.755	2.745

(d) (6 marks) The two models have quite different functional forms and coefficient estimates, but quite similar forecasts. Give an explanation for this.

(2 marks) The ARMA(1,1) model for Z_t has the general form

$$(1 - \phi_1 L)Z_t = (1 + \theta_1 L)U_t$$

in which the estimate for θ_1 is $\hat{\theta}_1 = 0.355$. Since this is less than 1 in magnitude, the moving average model is *invertible*.

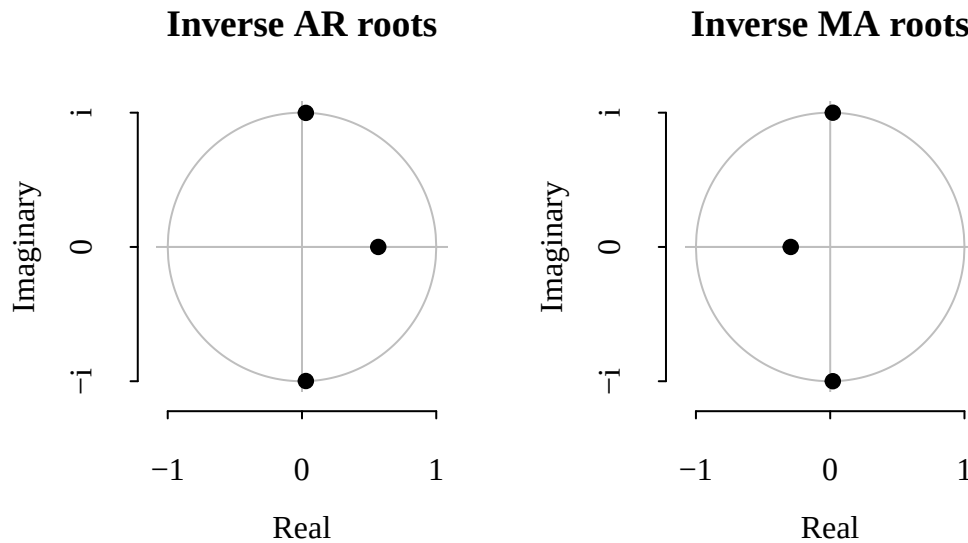
(2 marks) It follows that the ARMA(1,1) model can be expressed as a long autoregression

$$(1 - \phi_1^* L - \phi_2^* L^2 - \dots)Z_t = U_t$$

in which the coefficients ϕ_j^* do not need to be explicitly derived, but the invertibility implies their magnitudes decline towards zero as j increases.

(2 marks) Because of this, the ARMA(2,0) model, despite its short autoregressive length, is providing a quite close approximation to the ARMA(1,1) model, resulting in quite similar forecasts.

Referring to the table of AICc values, the ARMA(3,3) model may be another candidate model to try. The estimation of this model produces the following plot of its inverse roots.



(e) (5 marks) Give an explanation of what information these two plots suggest about the specification of the ARMA(3,3) model.

(2 marks) Both the AR and MA components of the show two unit roots and one stationary root. (It may be noted, although isn't required, that both AR and MA unit roots are occurring at/approximately $\pm i$.) The puzzling implication would be the AR component is non-stationary and MA component is non-invertible.

(2 marks) However these two sets of unit roots are also *equal*, implying they constitute *common factors* on either side of the ARMA(3,3) model. i.e. they cancel out, reducing to the preferred ARMA(1,1) model.

(1 marks) A lag polynomial representation is

$$(1 - \zeta_1^{-1}L)(1 - \zeta_2^{-1}L)(1 - \zeta_3^{-1}L)Z_t = (1 - \xi_1^{-1}L)(1 - \xi_2^{-1}L)(1 - \xi_3^{-1}L)U_t$$
 where $\zeta_2^{-1} = \xi_2^{-1} \approx$ and $\zeta_3^{-1} = \xi_3^{-1} \approx -i$ provide the common factors that cancel out.

If you run out of space for Question 3, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ Please tick this box if you used extra pages for any part of Question 3.

Question 4. (12 marks)

We have $n = 300$ observations on two time series, $Y_{1,t}$ and $Y_{2,t}$, to use as estimation sample, with the following $h = 5$ observations retained for forecast evaluation purposes. Following specification searches, the following models are fitted:

- univariate AR(1) model for $Y_{1,t}$,
- univariate AR(2) model for $Y_{2,t}$,
- bivariate VAR(1) model for $Y_{1,t}$ and $Y_{2,t}$.

Forecasts of $Y_{1,t}$ and $Y_{2,t}$ were generated from these three models for $t = n + 1, \dots, n + 5$, and Root Mean Squared Errors calculated. The following table shows the RMSEs for the forecasts of $Y_{1,t}$ (Y1) and $Y_{2,t}$ (Y2) from the corresponding univariate AR model (AR) and the bivariate VAR(1) model (VAR).

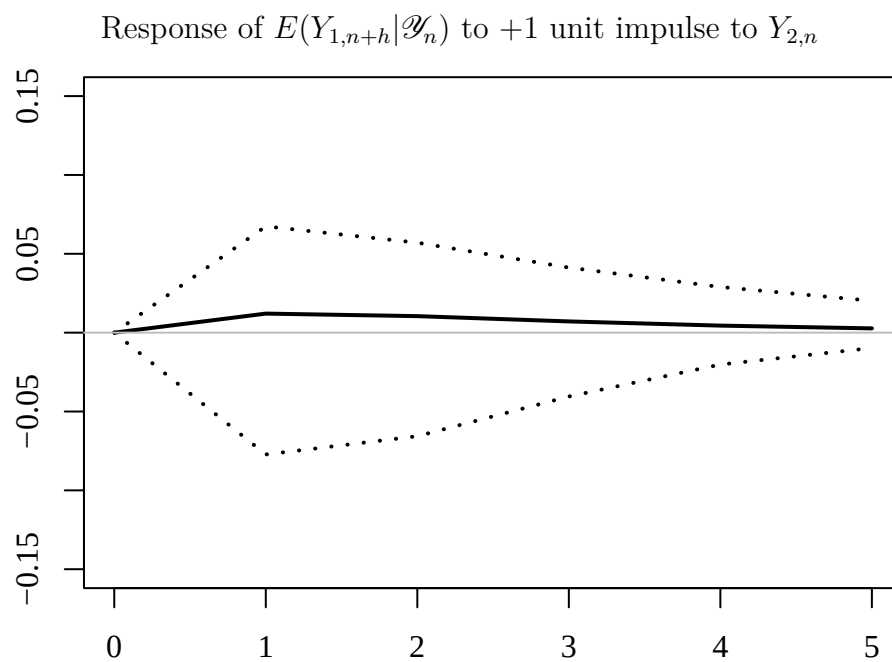
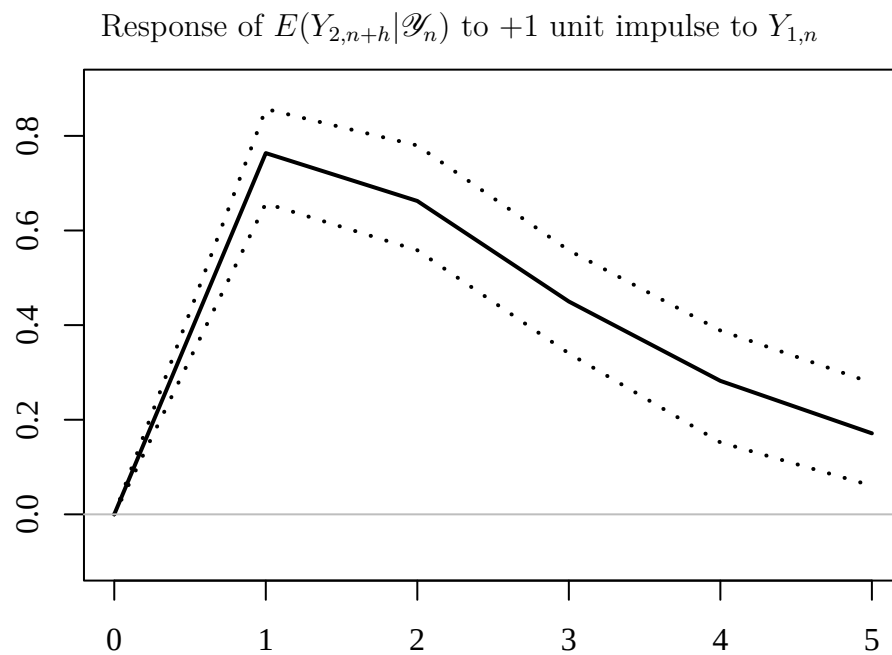
	Y1	Y2
VAR	0.734	0.908
AR	0.703	0.993

- (a) (4 marks) Explain what these results imply about the presence or absence of Granger causality in either direction between $Y_{1,t}$ and $Y_{2,t}$.

(2 marks) The RMSE for 'Y1' is higher for the bivariate VAR model than it is for the univariate AR model. Including 'Y2' in the model for forecasting 'Y1' makes the forecasts for 'Y1' worse, and certainly not improved, implying no Granger causality from 'Y2' to 'Y1'.

(2 marks) The RMSE for 'Y2' is lower for the bivariate VAR model than it is for the univariate AR model. Including 'Y1' in the model for forecasting 'Y2' improves the forecasts for 'Y2', implying Granger causality from 'Y1' to 'Y2'.

Using the VAR(1) model for $Y_{1,t}$ and $Y_{2,t}$, *non-orthogonalised* impulse response functions were calculated and plotted below, along with 95% confidence intervals (in dashed lines).



If you run out of space for Question 4, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ Please tick this box if you used extra pages for any part of Question 4.

(b) (4 marks) Interpret these impulse response functions.

(2 marks) The first impulse response function shows that a +1 unit impulse to $Y_{1,n}$ would result in significantly increased forecasts for $Y_{2,n+h}$ for at least the first 5 steps ahead. The forecast response is greatest for $h = 1$ and declines towards zero as h increases.

(2 marks) The second impulse response function shows that forecasts of $Y_{1,n+h}$ are barely revised at all in response to a +1 unit impulse to $Y_{2,n}$, being close to zero and statistically insignificant for every h .

(c) (4 marks) In addition discuss whether the implications of the impulse responses are consistent or inconsistent with the Granger causality findings in part (a).

(2 marks) The interpretation of the first impulse response function is consistent with the Granger causality finding of Granger causality from $Y_{1,t}$ to $Y_{2,t}$. If $Y_{1,t}$ is useful for forecasting future values of $Y_{2,t}$ then changes to $Y_{1,t}$ should translate into changes in the forecast for $Y_{2,t}$.

(2 marks) The interpretation of the second impulse response function is consistent with the Granger causality finding of *no* Granger causality from $Y_{2,t}$ to $Y_{1,t}$. If $Y_{2,t}$ is not useful for forecasting $Y_{1,t}$ then we would not expect that changes in $Y_{2,t}$ would produce changes in forecasts of $Y_{1,t}$.

Question 5. (14 marks)

Consider a model for a quarterly time series Y_t :

$$Y_t = \beta_0 + \beta_1 Q_{1,t} + \beta_2 Q_{2,t} + \beta_3 Q_{3,t} + Z_t$$

$$Z_t = 1.3Z_{t-1} - 0.4Z_{t-2} + U_t + 0.5U_{t-1},$$

where $Q_{1,t}$, $Q_{2,t}$ and $Q_{3,t}$ are seasonal dummy variables taking the value 1 in quarters 1, 2 and 3 respectively, and the value 0 otherwise.

(a) (2 marks) Express the equation for Z_t using lag operators and polynomials.

(2 marks) $(1 - 1.3L + 0.4L^2)Z_t = (1 + 0.5L)U_t$

(b) (4 marks) Is it possible that the model for Z_t is stationary? Explain why or why not, and why the answer needs to be qualified by "is it possible".

(1 marks) The autoregressive lag polynomial can be factored as

$$(1 - 0.8L)(1 - 0.5L).$$

(1 mark) This implies inverse roots of 0.8 and 0.5, both inside the unit circle.

(1 mark) Because of this, it is possible that the AR model is stationary...

(1 mark) ...provided appropriate assumptions about the mean and variance of the initial values are made to ensure the mean and variance of Z_t are constant for all t . (No need to try to derive these.)

(c) (2 marks) Is the model for Z_t is invertible? Explain why or why not.

(1 mark) The MA lag polynomial $(1 + 0.5L)$ has inverse root 0.5, which is inside the unit circle.

(1 mark) Therefore it is invertible.

(d) (2 marks) Does the model for Z_t contain a common factor? Explain why or why not.

(2 marks) The representation

$$(1 - 0.8L)(1 - 0.5L)Z_t = (1 + 0.5L)U_t$$

reveals no common factors, i.e. no cancellation of any of terms from AR and MA components.

(e) (4 marks) Is it possible that the entire model for Y_t (i.e. both equations considered together) is stationary? If necessary, explain any special case(s) that affect your general answer.

(2 marks) In general Y_t will be non-stationary because the seasonal dummies have the effect of producing a different mean $E(Y_t)$ depending on the season.

(2 marks) In the special case of $\beta_1 = \beta_2 = \beta_3 = 0$ then Y_t can be stationary because then it has constant mean β_0 and otherwise inherits the stationarity properties of Z_t in part (b).

If you run out of space for Question 5, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

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Please tick this box if you used extra pages for any part of Question 5.

Question 6. (16 marks)

Let Y_t be a time series that satisfies the following AR(1)-ARCH(1) model

$$Y_t = \beta_0 + Z_t$$

$$Z_t = \phi_1 Z_{t-1} + U_t$$

$$E(U_t | \mathcal{Y}_{t-1}) = 0$$

$$E(U_t^2 | \mathcal{Y}_{t-1}) = \omega + \alpha_1 U_{t-1}^2.$$

- (a) (2 marks) Is it valid to consider U_t as the *one-step-ahead prediction error* in this model when it has the ARCH(1) structure?

(2 marks) It is valid because $E(U_t | \mathcal{Y}_{t-1}) = 0$ implies U_t is not 1-step-ahead predictable. The conditional variance structure does not affect this.

- (b) (2 marks) Why is it required to impose $\omega > 0$ and $\alpha_1 >$ in this model?

(2 marks) It is necessary for the model to imply positive values for $E(U_t^2 | \mathcal{Y}_{t-1})$, and the imposition of $\omega > 0$ and $\alpha_1 >$ (along with the squared in U_{t-1}^2) guarantees this.

(c) (6 marks) What parameter restriction is required for U_t to be stationary? Under this restriction, derive the *unconditional* mean and variance of U_t .

(1 mark) The stationarity condition for U_t is $\alpha_1 < 1$ as will become evident.

(1 marks) The unconditional mean follows from the LIE:

$$E(U_t) \stackrel{\text{LIE}}{=} E(E(U_t | \mathcal{Y}_{t-1})) = 0$$

(2 marks) The unconditional variance follows from

$$\begin{aligned} \text{var}(U_t) = E(U_t^2) &\stackrel{\text{LIE}}{=} E(E(U_t^2 | \mathcal{Y}_{t-1})) = E(\omega + \alpha_1 U_{t-1}^2) \\ &= \omega + \alpha_1 E(U_{t-1}^2) \end{aligned}$$

(1 marks) Stationarity requires $E(U_t^2) = E(U_{t-1}^2)$ for all t , so that

$$E(U_t^2) = \omega + \alpha_1 E(U_t^2)$$

(1 marks) which solves to give

$$E(U_t^2) = \frac{\omega}{1 - \alpha_1},$$

which requires $\alpha_1 < 1$ to be valid.

- (d) (4 marks) Give expressions for the conditional mean and conditional variance of Y_t , i.e. $E(Y_t | \mathcal{Y}_{t-1})$ and $\text{var}(Y_t | \mathcal{Y}_{t-1})$, as functions of lags of Y_t .

(2 marks) Substitution of $Z_t = Y_t - \beta_0$ gives

$$Y_t = \beta_0 + \phi_1(Y_{t-1} - \beta_0) + U_t$$

and hence

$$E(Y_t | \mathcal{Y}_{t-1}) = \beta_0 + \phi_1(Y_{t-1} - \beta_0)$$

since $E(U_t | \mathcal{Y}_{t-1}) = 0$.

It also implies that

$$\text{var}(Y_t | \mathcal{Y}_{t-1}) = \text{var}(U_t | \mathcal{Y}_{t-1}).$$

(2 marks) Substitution for U_{t-1} , Z_{t-1} and Z_{t-2} gives

$$U_{t-1} = (Y_{t-1} - \beta_0) - \phi_1(Y_{t-2} - \beta_0)$$

so that

$$\begin{aligned} \text{var}(Y_t | \mathcal{Y}_{t-1}) &= \text{var}(U_t | \mathcal{Y}_{t-1}) = \omega + \alpha_1 U_{t-1}^2 \\ &= \omega + \alpha_1 ((Y_{t-1} - \beta_0) - \phi_1(Y_{t-2} - \beta_0))^2 \end{aligned}$$

- (e) (2 marks) How would such expressions for $E(Y_t | \mathcal{Y}_{t-1})$ and $\text{var}(Y_t | \mathcal{Y}_{t-1})$ be useful for constructing a normal 95% prediction interval for the one-step-ahead forecast $E(Y_{n+1} | \mathcal{Y}_n)$?

(2 marks) The normal 95% prediction interval formula is

$$\hat{E}(Y_{n+1} | \mathcal{Y}_n) \pm 1.96 \times \sqrt{\widehat{\text{var}}(Y_{n+1} | \mathcal{Y}_n)}$$

Expressions for the conditional mean and variance therefore provide the formulae that are used to compute the prediction interval once the parameters are estimated.

(f) (2 marks) Referring to your answer to (d), which lags of Y_n are used in computing the prediction interval in the AR(1)-ARCH(1) model in this question?

(2 marks) Having estimated the model parameters, we would have

$$\begin{aligned}\hat{E}(Y_{n+1}|\mathcal{Y}_n) &= \hat{\beta}_0 + \hat{\phi}_1(Y_n - \hat{\beta}_0) \\ \widehat{\text{var}}(Y_{n+1}|\mathcal{Y}_n) &= \hat{\omega} + \hat{\alpha}_1((Y_n - \hat{\beta}_0) - \hat{\phi}_1(Y_{n-1} - \hat{\beta}_0))^2\end{aligned}$$

The prediction interval for Y_{n+1} therefore depends on Y_n and Y_{n-1} .

(g) (2 marks) Referring to your answer to (d), how does value of the one-step-ahead forecast for Y_{n+1} relate to the value of the ARCH(1) coefficient (α_1)?

(2 marks) $\hat{E}(Y_{n+1}|\mathcal{Y}_n) = \hat{\beta}_0 + \hat{\phi}_1(Y_n - \hat{\beta}_0)$

The point forecast does not depend on the ARCH(1) coefficient at all.

(h) (2 marks) Referring to your answer to (d), how does the width of the one-step-ahead prediction interval for Y_{n+1} relate to the value of the ARCH(1) coefficient (α_1)?

(2 marks) The width of the prediction interval is

$$2 \times 1.96 \times \sqrt{\widehat{\text{var}}(Y_{n+1}|\mathcal{Y}_n)}$$

and the expression in (d) for the conditional variance shows that it is increasing in the value of the ARCH(1) coefficient. i.e. the width of the prediction interval has a positive relationship with the ARCH(1) coefficient, all else equal.

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